EMERGENCY BLOOD MANAGEMENT PLAN FOR SHORTAGES OF BLOOD AND BLOOD PRODUCTS – ROYAL DEVON UNIVERSITY HEALTHCARE NHS FOUNDATION TRUST

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1. Purpose and background

1.1 NHS emergency planning requires the development of contingency plans to ensure the effective use of available blood components when stocks fall to specific trigger levels. Pre-determined plans are critical to ensure transfusion support remains available for the patients who need it most.

1.2 The National Health Blood and Transplant (NHSBT) service has a plan for managing these shortages from the collection, processing and supplying perspectives.

1.3 The contingency planning group of the National Blood Transfusion Committee required hospitals to develop emergency blood management arrangements to manage blood component shortages in a variety of situations, including:

Short term shortages, for example, during bad weather, an influenza outbreak, unexpected increases in demand or security issues which prevent donors coming forward to donate blood.

Prolonged shortages which could result from a number of circumstances, for example the introduction of further measures to reduce the risk of disease transmission by transfusion or changes in processing. The COVID-19 pandemic also prompted concerns around component shortages that may be prolonged.

1.4 The overall NHSBT and local strategy plans have the following two key aims:

- Blood components are available for essential transfusions
- Overall component usage is reduced to ensure supply remains available for essential transfusions

1.5 Shortages may include all blood components, or may be component specific.

1.6 The Royal Devon University Healthcare NHS Foundation Trust emergency blood management plan described in this document is focussed mainly on red cell component shortages. This local plan applies to the eastern and northern sites.

2. Emergency Blood Management Plan for shortages of blood components at the Royal Devon University Healthcare NHS Foundation Trust

2.1 The four phases of the plan

The plan is structured to provide a framework of actions for NHSBT and hospitals at four phases

Green: Normal circumstances where supply meets demand
Pre-Amber: Reduced availability of blood for short or prolonged period
Amber: Reduced availability of blood for a short or prolonged period with impact on clinical activity
Red: Severe, prolonged shortages with impact on clinical activity

2.1.1 A schematic of the four phases and actions required is in appendix I.

2.1.2 To simplify the management of patients during blood component shortages a table stating transfusion requirements of three categories has been created for each component type (appendix II, III and IV). This is to assist hospitals with prioritising patients to achieve the required reduction in red cell usage. It is recognised that clinical judgement is an essential part of decision-making for individual patients.

2.1.3 A check list is available that can be used to support the actions required during blood component shortages (appendix V)

2.2 The Emergency Blood Management Group

2.2.1 The Emergency Blood Management Group (EBMG) for the hospital will be called upon in the event of an amber or red alert to attend a meeting to discuss the action to be taken and any measures that need to be put in place to conserve the amount of blood components in the Trust. The EBMG consists of the following individuals (where the post holder is unable to attend a nominated representative should attend the meeting):

Emergency Blood Management Plan Group
Medical Director or deputy
Care Group Deputy Medical Director or deputy– chair of the meeting
Director of Nursing
Director of Operations
Clinical Director, Critical Care/Lead Clinical Anaesthetist
The Haematologist responsible for Transfusion/Transfusion Doctor
Consultant surgeon
Consultant physician
Consultant anaesthetist
Consultant obs/gynae

Blood transfusion laboratory manager
Pathology Service Manager
Specialist Services Divisional Director
Administrative support
Communication team representative

2.2 Action for Green Phase

2.2.1 Obtain senior management and NHS Trust Board commitment.

2.2.2 Ensure that effective blood transfusion policies for the appropriate use of red cells are in place, implemented and monitored.

2.2.3 This is the normal operation of the hospital transfusion team HTT) and patient blood management group (PBMG), supporting appropriate use of blood components according to national and local guidelines.

2.3 Pre-amber phase

2.3.1 Ensure EBMP arrangements are in place and that the EBMP group can be convened quickly if needed. This is in anticipation of a potential Amber alert should the situation not improve. The Chief Medical Officer must be alerted at the potential move to Amber and the implications of this.

2.3.2 Patient blood management tools utilised within the Trust ensure that transfusions are appropriate, these are monitored by the HTT and PBMG.

2.3.3 Reduce stockholding of red cells on both sites if possible. The stocks and usage will be reviewed by the transfusion management team and stocks will be reduced as determined by review.

2.3.4 Red cell stock levels will be shared with the NHSBT via the VANESA site on a daily basis. Consideration will be given to reduction of stocks with more frequent stock ordering and use of ad hoc deliveries.

2.4 Action for Amber Phase (in addition to pre-amber actions)

2.4.1 If national stocks fall to less than 2 days, or an imminent threat to the blood supply is identified, NHSBT will move to AMBER phase, following a pre-amber warning to hospitals.
Continuation of elective surgery will depend on blood stock levels and the urgency of the surgery (appendix VI). The EBMG will convene to discuss the current situation within the hospital and any surgical cases for potential postponement.

2.4.2 Communication will be sent out to all Trust staff via HUB (see appendix

VII for example of communication).

2.4.3 An example agenda for the meetings is provided in Appendix VIII.

2.4.4 Blood component stocks held in the eastern and northern sites will be reviewed and further reduced if appropriate. Blood components may be transferred across sites to reduce risk of wastage.

2.4.5 Consideration may be given to reducing the transfusion trigger for transfusions if agreed regionally.

2.4.6 All orders for non-bleeding adult patients, transfusion is outside of the national indication codes for transfusion[1] are referred to the Haematology advice and guidance doctor for approval or discussion with relevant clinical team. Records are kept of referrals and discussions. The haematology advice and guidance doctor's remote clinic will be cancelled to allow enough capacity to undertake this work. If the advice and guidance doctor is away, the requests will be forwarded to the on-call haematology team during normal working hours (9am to 5pm). Requests outside of this time will not be screened. NOTE: review of check and challenge performed for 2024 amber shortage showed minimal savings and was timing consuming for clinical and laboratory staff. The current clinical decision support in EPIC appears to be effective in avoiding unnecessary transfusion. Agreement on implementation of check and challenge should be made by the group as an optional approach.

2.4.7 Reduction of the reservation period for blood to 12 hours wherever possible.

2.4.8 Action for major trauma patients during an amber phase if agreed across the region:

- Use a plasma-first approach and give 2 to 4 units fresh frozen plasma (FFP) as first transfusion product
- Use an inverse ratio of FFP to Blood (2 FFP to 1 RBC). (FFP:RBC ratios may be flexed between 3:1 and 1:3 depending on supply, as guided by local and regional transfusion services)
- If platelet supply is restricted, give the first pool of platelets at 8 RBC units and every 8 units of RBCs thereafter

2.4.9 Consideration should be given to use of the 60-minute temperature excursion rule in accordance with JPAC guidance (JPAC 2016), this will be performed on a case by case basis by the transfusion laboratory management team.

2.4.10 The medical director or deputy will raise at the Integrated Care Board

[1] NBTC indication codes - Final - Jan 20 (004).pdf

(ICB) or equivalent to ensure a regionally agreed response to the shortage - particularly around deferral of surgery and changes to major haemorrhage protocols.

2.4.11 A Peninsular pathology network group will be convened to ensure regional approach from the laboratory managers. This will be attended by the transfusion laboratory manager or deputy.

2.4.12 Cell salvage will be introduced during the blood shortage period for all LSCS.

2.4.13 Cell salvage 'suck and save' will be introduced for other surgical cases where there is an expectation that adequate collection of blood could be utilised for re-infusion.

2.4.14 Patients scheduled for AAA will be clinically assessed through MDT to be low risk for blood loss or those at increased risk of rupture (P2 priority and AAA>7cm). All pre-operative optimisation (cell salvage, Hb optimisation, use of IV iron etc) will be included in decision making for surgical intervention.

2.4 Action for Red Phase (in addition to amber actions)

2.4.1 Reduce stockholding to the level agreed with the NHSBT.

2.4.2 Reduce usage to the categories indicated by NHSBT (appendix I).

2.4.3 Daily review of the blood shortage and its impact on patient care by the EBMG.

2.4.4 Medical assessment of all requests by a Consultant Haematologist.

2.4.5 An order of priority based on clinical need as determined and agreed by the EBMG.

2.4.6 The establishment of links between hospitals to utilise regional stocks more effectively as appropriate and agreed by the Peninsular pathology network group.

2.4.7 In cases of actual or potential massive blood loss a Consultant Haematologist must be contacted by the referring clinical team to allow discussion and planning of patient management and blood product provision.

2.4.8 Action for major trauma patients during a red phase if agreed as a regional response:

- Repeatedly assess futility during major haemorrhage resuscitation
- At start of a major haemorrhage resuscitation, and after each 8 units of RBCs, assess the patient's status and consult with lead clinicians within the resuscitative/operative teams about continued transfusion therapy and overall futility
- Involve consultant in transfusion medicine in decision-making as early as possible
- Continue haemostatic measures in the Amber phase.
- After each 8 units of RBCs, additionally give a therapeutic dose of both prothrombin complex concentrate (1000iu) and fibrinogen concentrate (4g). Cryoprecipitate 2 pools may be used instead of fibrinogen concentrate if not available

3 Recovery from shortage

3.1 The NHSBT will send an email informing the transfusion laboratory that stocks have risen to a level where hospitals can move to Amber or Green phase.

3.2 The transfusion laboratory manager or deputy will disseminate the information as above.

3.3 The hospital should ensure that immediate demand does not return the national stocks to below critical levels by using a phased return to normal stock levels. The return to normal activity levels should similarly be phased in particular, elective surgery backlogs should not be compressed into the immediate post recovery period.

3.4 The EBMG will convene at the earliest opportunity to review the effect of the blood shortage and amend the blood shortage arrangements as necessary. Any recommendations will be fed back to the Patient Blood Management Group.

4 Platelet, Fresh Frozen Plasma and Cryoprecipitate Shortages

4.1 When a shortage of the above components is encountered, the normal pathway of notification will be carried out so that all clinicians are informed.

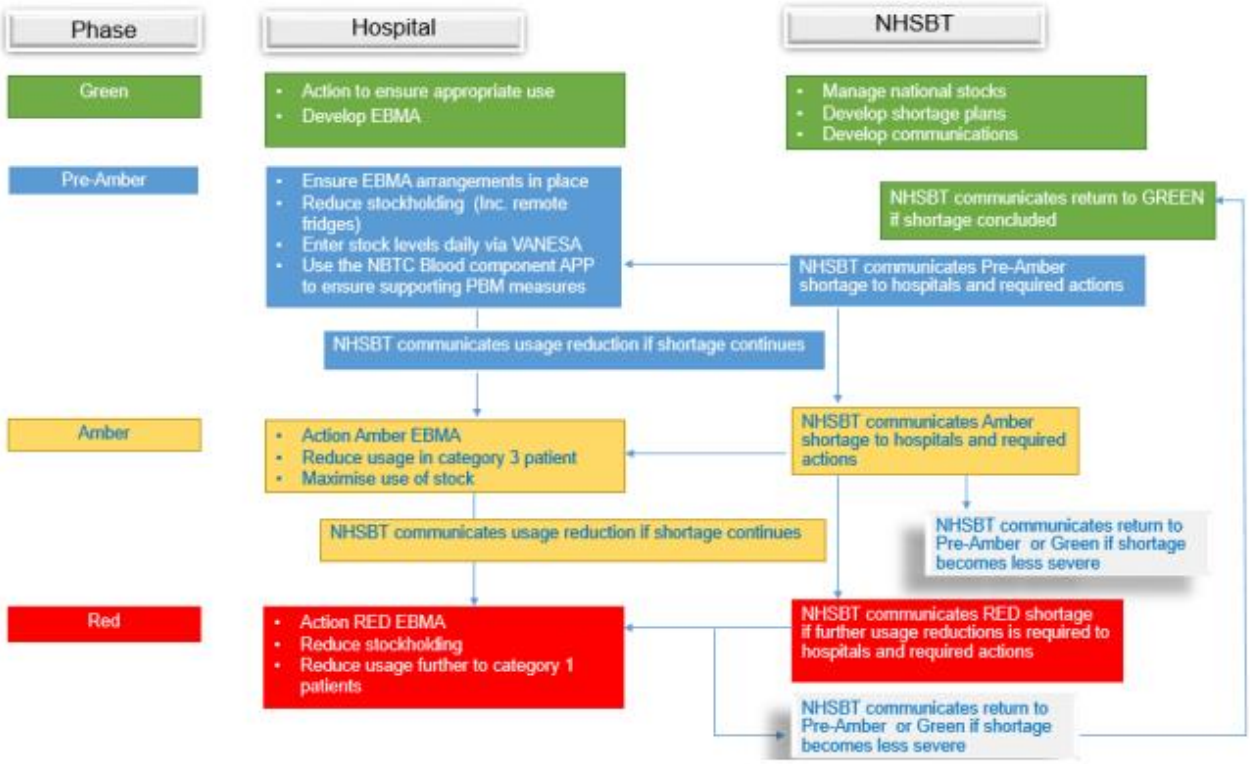
4.2 Requests should be made to the blood transfusion department giving as much information as possible, these will be referred to a Consultant Haematologist.

4.3 The decision to supply the blood component will be taken by the Consultant Haematologist after discussion with the requesting Consultant.

4.4 Guides for the prioritisation of patients during component shortages are in appendix III and IV.

4.6 Instructions and recommendations for the use of platelet concentrate components will be provided by the NHSBT.

Appendix I: Schematic of red cell shortage plan



Appendix II Guide to the prioritisation of patients for red cells

Guide to the prioritisation of patients for red cells

Category 1 These patients will remain highest priority of transfusion	Category 2 These patients will be transfused in the amber but not the red phase	Category 3 These patients will not be transfused in the amber phase
Resuscitation Resuscitation of life-threatening /on-going blood loss including trauma		
Surgical support Emergency surgery* including cardiac and vascular surgery**, and organ transplantation Cancer surgery with the intention of cure	**Surgery/obstetrics** Cancer surgery (palliative) Symptomatic but not life- threatening post-operative or post-partum anaemia Urgent*** surgery	**Surgery** Elective surgery which is likely to require donor blood support
Non-surgical anaemias Life-threatening anaemia including patients requiring in-utero support and high dependency care/SCBU Stem cell transplantation, or chemotherapy **** Severe bone marrow failure Transfusion-dependent anaemias including thalassaemia and myelodysplasia Sickle cell disease (SCD) patients on regular transfusion programmes for prevention of complications of SCD Organ transplant	**Non-surgical anaemias** Symptomatic but not life-threatening anaemia	

* Emergency patient likely to die within 24 hours without surgery.
** With the exception of poor risk aortic aneurysm patients who rarely survive but who may require large volumes of blood.
*** Urgent - patient likely to have major morbidity if surgery not carried out.
**** Planned stem cell transplant or chemotherapy may be deferred if possible.

Appendix III Guide to the prioritisation of patients for platelets

Category 1 **Patients to be treated in red phase**	Category 2 **Patients to be treated in red and amber phases**	Category 3
Massive haemorrhage and critical care Massive transfusion for any condition including obstetrics, emergency surgery and trauma, with on-going bleeding, maintain > 50 x 10^9/L Aim for >100 x 10^9/L if multiple trauma or CNS trauma Bleeding in the presence of sepsis/acute DIC, maintain >50 x 10^9/L	**Critical care** Patients resuscitated following massive transfusion with no on-going active bleeding, maintain > 50 x 10^9/L Surgery: emergency surgery but not urgent surgery for a patient requiring platelet support Transfusion triggers for invasive procedures According to BSH guidelines	**Surgery** Elective, non-urgent surgery likely to require platelet support for thrombocytopenia or congenital/acquired platelet defects
Bone marrow failure Active bleeding associated with severe thrombocytopenia or functional platelet defects Immune thrombocytopenia if serious life-threatening bleeding	**Bone marrow failure** All other indications except those in category 1 or 3	***Bone marrow failure** Prophylactic transfusion of stable patients following autologous stem cell transplant
Neonates For preterm neonates with very severe thrombocytopenia (platelet count below 25 × 10^9/l) platelet transfusions should be administered in addition to treating the underlying cause of the thrombocytopenia. Suggested threshold counts for platelet transfusions in other situations are given in the BSH guidelines.		

** Prophylactic transfusion category should include WHO grade 1 bleeding (as in TOPPS trial). Exclusions – previous WHO > grade 3 bleed, inherited haemostatic or thrombotic disorder, requirement for therapeutic doses of anticoagulation, acute promyelocytic leukaemia, prior to surgery/invasive procedure.*

Appendix IV Guide to the prioritisation of patients for plasma

Category 1 These patients will remain highest priority of transfusion	Category 2 These patients will be transfused in the amber but not the red phase	Category 3 These patients will not be transfused in the amber phase
Massive haemorrhage and critical care Massive transfusion for any condition including obstetrics, emergency surgery and trauma, with on-going bleeding Bleeding in the presence of sepsis/acute DIC	**Surgery/obstetrics** Cancer surgery (palliative) Urgent*** surgery	**Invasive procedure** Elective procedures which are likely to require plasma support **Transfusion triggers for invasive procedures** According to BSH guidelines
Surgical support Emergency surgery* including cardiac and vascular surgery**, and organ transplantation Cancer surgery with the intention of cure	**Plasma exchange** Conditions requiring urgent treatment	**Surgery** Elective surgery which is likely to require plasma support
Plasma exchange Life-threatening conditions requiring urgent treatment (e.g. TTP)		
Neonates For preterm neonates with very severe coagulopathy and bleeding		
Bone marrow failure Active bleeding associated with severe coagulopathy		

* *Emergency – patient likely to die within 24 hours without surgery.*
** *With the exception of poor risk aortic aneurysm patients who rarely survive but who may require large volumes of blood.*
*** *Urgent – patient likely to have major morbidity if surgery not carried out.*

Appendix V Emergency blood management arrangement checklist

Checklist: Emergency Blood Management Arrangements

NHS Blood and Transplant

This guidance has been developed in conjunction with the National Blood Transfusion Committee (NBTC) red cell, platelet and plasma shortage plans and aims to create a short and concise series of steps to follow in the case of shortage.

Click on the white boxes to tick each step

Checklist for green

This is the business as usual phase of the EBMA

Clinical teams to ensure:

1. your EBMA plan is up to date ☐
2. the members of the Emergency Blood Management (EBM) Group are aware of the plan ☐
3. PBM strategies such as anaemia treatment, cell salvage and adherence to national indication codes are followed ☐
4. familiarity with trust Emergency Preparedness Resilience and Response (EPRR) plans and command structures ☐
5. communications are drafted for use if a move to amber/red is required ☐
6. confirm stocks of Anti D, Tranexamic acid, Fibrinogen, Albumin, Lyoplas, Octaplas and Desmopressin - ensure process to order additional stocks is established ☐
7. process agreed for the review of appropriateness of blood requests with haematology clinicians as needed ☐

Checklist for pre-amber:

1. Ensure EBMA arrangements in place ☐
2. Reduce stockholding (inc. remote fridges) ☐
3. Transfusion staff to enter stock levels daily via VANESA ☐
4. Clinical staff to use the NBTC Blood component APP to ensure supporting PBM measures ☐

Checklist for amber

NHSBT will inform transfusion team that amber stage needs to be actioned if stock falls below 2 days for red cells, 5 days for plasma or 1 day for platelets.

General:

1. Activate EBMA and convene EBM group ☐
2. Prepare to report stock levels and decisions made by EBM group for escalation trust-wide ☐
3. Arrange trust-wide communications (e.g. screensavers, emails, newsletters) ☐
4. Review satellite fridges; is stock still required? ☐
5. Consider pharmaceutical alternatives in appropriate patients with EBM group and disseminate decision ☐
6. Contact areas where transfusions may stop ☐
7. Clinical staff to reprioritise prophylactic transfusions ☐

Red cells: Consider:

1. are all PBM methods being used, review scale up? ☐

Platelets: Consider:

1. D positive platelets for D negative patients (with anti-D) ☐

Plasma: Consider:

1. conserving AB plasma for group AB patients ☐

Checklist for red

The move to red phase will be communicated to trusts if there are severe shortages of either red cells, plasma or platelets.

Complete all amber actions.

General:

1. Launch rota for senior haematology clinicians, used to support the transfusion laboratory and triage requests ☐
2. Update communications to reflect change to red phase ☐
3. Remove all stock from satellite fridges except emergency group O from acute areas e.g. ED and maternity ☐
4. Contact clinical areas where transfusions will not take place. ☐
5. Requests should be vetted by Blood Transfusion Laboratory staff with support from Blood Transfusion Consultant. ☐

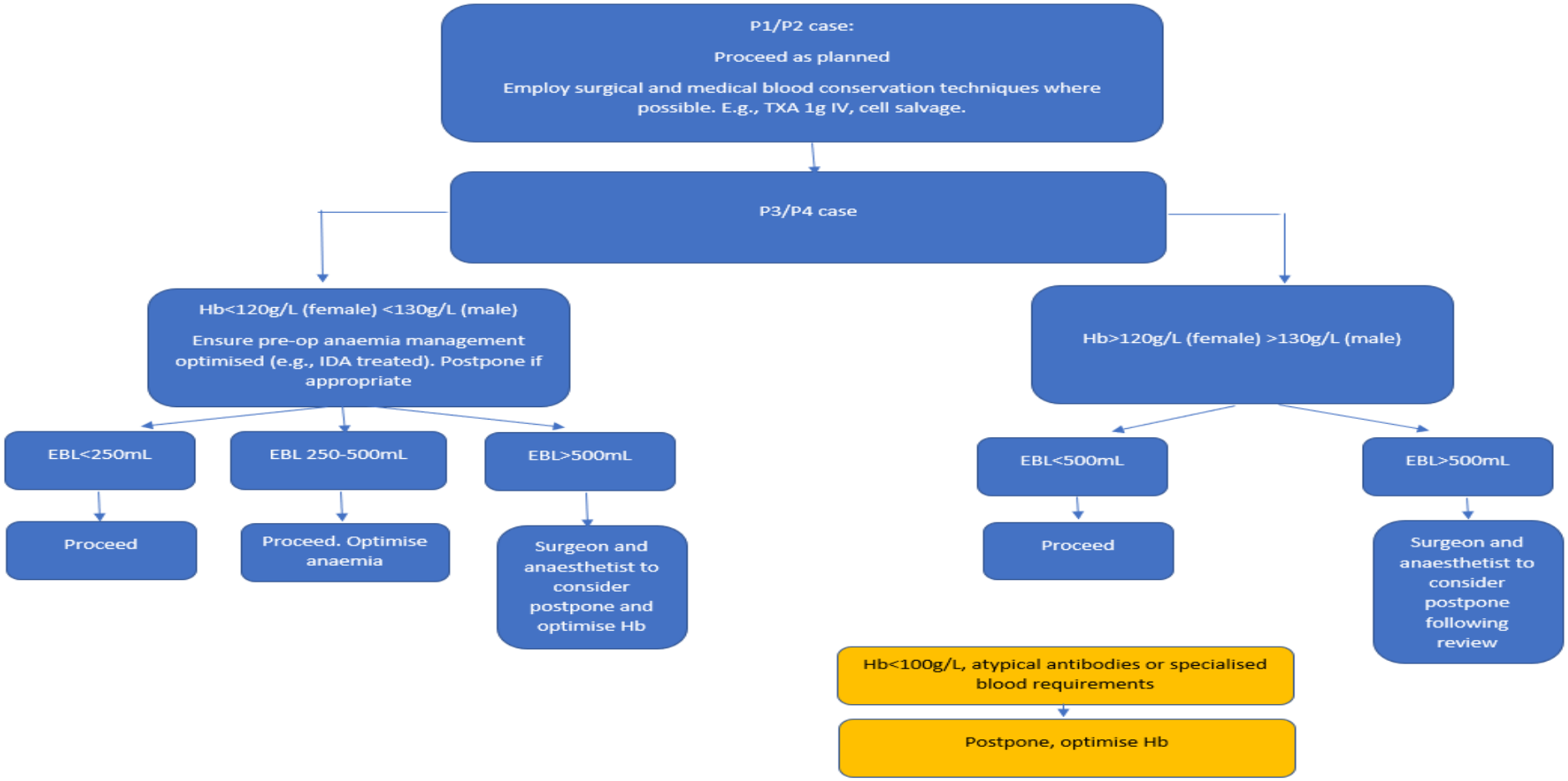
Recovery phase:

NHSBT will inform the transfusion team of return to 'green' phase.

1. Convene the EBM group ☐
2. Ensure that change in clinical activity reflects blood stock levels ☐
3. Use trust-wide communications to update staff ☐

CLICK HERE for more information

Appendix VI Flow chart for surgical decision making



Appendix VII Example of communication to be sent to all Trust staff

National amber alert in place on blood stocks: please read for actions

The NHS Blood and Transplant Service (NHSBT) has issued an amber blood shortage warning for *(add component type and blood group as relevant)* meaning that stock nationally has fallen below two days.

During this time, we need to work together to ensure that blood stocks are conserved for those who need them most.

In response, the Trust Emergency Blood Management Group has met to discuss the implications of this and the actions needed to support life-saving transfusions during this alert period. Please see HUB for full details.

Thank you to colleagues who expressed an interest in donating blood in response to the shortage and have contacted us to ask about a local collection. The NHSBT have confirmed that their mobile teams are currently at full capacity so unable to provide any additional local collections.

Appointments are available at NHSBT town and city donor centres so please help share the blood donor appeal message widely with friends and family. Donations are needed regularly to maintain stocks so booking a future appointment is just as helpful.

If you wish to donate, we suggest trying the app, as there are reports of long queues on the phone lines: The NHSGiveBlood app - NHS Blood Donation

Appendix VIII Example agenda for meeting

Item	Item	Lead	Action	Time
1 1.2	Welcome & Apologies for absence: Please note the meeting will be recorded for Minute taking purposes.			10:00
2	Minutes of last meeting –		To Note	
3	Matters arising & action list		To Note	
4	Amber actions outstanding			
4.1	• Regional approach		Information/ Discussion	
5	**Stocks update**			
5.1 5.2	• https://hospital.blood.co.uk/business-continuity/blood-stocks/ • Local blood donation session		Information/ Discussion	
6	**Cases to discuss**			
6.1	Elective surgeries for patient likely needing transfusion		Information/ Discussion	

7	**Any other Business**			
7.1	• Amber Alert Support: national resources		Information/ Discussion	
20	**Date of Next Meeting**			
20.1		ALL	To Note	

References

JPAC - Joint UKBTS Professional Advisory Committee 2016: https://www.transfusionguidelines.org/uk-transfusion-committees/national-blood-transfusion-committee/responses-and-recommendations

National Blood Transfusion Committee A Plan for NHS Blood and Transplant and Hospitals to Address Red Cell Shortages July 2022 https://www.transfusionguidelines.org/uk-transfusion-committees/national-blood-transfusion-committee/responses-and-recommendations